Nonnegative and positive factorizations

Infinitesimal rigidity detects unique size-two factors in the presence of zeros

Difficulty: challenging **Importance:** interesting to specialist

Status: explicitly conjectured in the August 2026 source; checked 2026-09-08.

Context and notation

Write $\mathbb{S}_+^k$ for the cone of real symmetric positive semidefinite $k \times k$ matrices. For an entrywise nonnegative matrix $M \in \mathbb{R}_+^{p\times q}$, its **real positive semidefinite rank** is

$$\operatorname{rank}_{\mathrm{psd}}(M) = \min\{k \geq 1 : \exists A_1, \dots, A_p, B_1, \dots, B_q \in \mathbb{S}_+^k, \quad M_{ij} = \operatorname{tr}(A_i B_j) \text{ for all } i, j\}.$$

This definition concerns a family of matrix factors indexed by the rows and columns of M.

Problem statement

Let $M \in \mathbb{R}_+^{p\times q}$ satisfy $\operatorname{rank}(M) = 3$ and $\operatorname{rank}_{\mathrm{psd}}(M) = 2$, and fix any size-two factorization $M_{ij} = \operatorname{tr}(A_i B_j)$. Zero entries of M are allowed.

Call a collection of real symmetric matrices (E_i, F_j) a feasible linear direction if

$$\operatorname{tr}(E_i B_j) + \operatorname{tr}(A_i F_j) = 0 \quad \text{for every } i, j$$

and some $h > 0$ satisfies $A_i + tE_i \succeq 0$, $B_j + tF_j \succeq 0$ for all i, j and $t \in [0, h)$. The trace constraint is imposed only to first order; the perturbed factors need not represent M at positive t.

Question

Are the following conditions equivalent for every such factorization?

1. Every feasible linear direction has the form $E_i = dA_i$, $F_j = -dB_j$ for one real scalar d common to all factors.
2. Every other size-two factorization $(\widetilde{A}_i, \widetilde{B}_j)$ of M satisfies $\widetilde{A}_i = S^{\mathsf{T}} A_i S$ and $\widetilde{B}_j = S^{-1} B_j S^{-\mathsf{T}}$ for one $S \in GL(2, \mathbb{R})$.

This asks whether an infinitesimal test certifies uniqueness of the optimal matrix factors up to changes of basis. Condition 1 is exactly the source's 2-infinitesimal rigidity: for size two, the relevant second-degree Taylor polynomials are the complete principal minors, and Theorem 2.2 identifies the trivial directions as the common scalings above.

Reference and status

K. Dawson, S. Hoşten, K. Kubjas, and L. Metsälampi, *Uniqueness of size-2 positive semidefinite matrix factorizations*, v2, 31 August 2026, Definition 1, Remark 1, Theorem 2.2, and the conjecture immediately after Theorem 5.2. Theorem 5.2 proves the assertion when every entry of M is strictly positive. Lemma 19 identifies local and global rigidity in the stated rank class, so omitting the equivalent local condition does not weaken the source's remaining conjecture. Searches for the title, PSD rigidity with zeros, and later work found no resolution beyond this latest version.